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Real-time navigation of mecanum wheel-based autonomous mobile robot for a human-robot collaboration system

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Abstract: Autonomous Mobile Robots (AMRs) are utilized for handling goods and workpieces in factories and warehouses. The advantages of AMRs are completing automated tasks, reducing labour costs, improving efficiency and increasing convenience. In order to enable AMRs to navigate and move faster in all directions, this study focuses on the implementation of an AMR equipped with four Mecanum wheels to execute path planning tasks for a human-robot collaboration work. To achieve the path planning tasks, the AMR employs a Simultaneous Localization and Mapping (SLAM) algorithm based on the Robot Operating System (ROS) and light detection and ranging (LIDAR) sensors are used for mapping. With omnidirectional mobility of the AMR, a global path planning method is crucial to ensure the AMR's navigation in the environment. This study adopts the A* algorithm and Voronoi diagram as the global path planning method; it can find the shortest path for navigation of the AMR. To improve the efficiency of AMR path planning, the Voronoi diagram is used to describe the map and the shortest path is built by Dijkstra algorithm. The collision-free A* algorithm will avoid collisions effectively and find the optimal path according to the shortest path tree. Voronoi algorithm is more suitable for use in factory environments with obstacles. Data fusion technology significantly improves positioning accuracy through the integration of visual odometers, IMUs, and traditional odometers.

1. Introduction

With the rapid development of intelligent manufacturing and industrial automation technology, autonomous mobile robots (AMRs) are playing an increasingly critical role in factories and warehousing. The AMRs can autonomously complete tasks such as object handling and environmental navigation without human intervention. They can not only reduce labour costs effectively and improve operation efficiency, but also reduces the safety risks and operation errors by manual operations greatly. Therefore, the development of AMR systems with high mobility and high navigation accuracy has become an important direction for automatic handling operations in smart factories. Path planning is a critical component to enable efficient navigation of AMRs through complex environments with avoiding obstacles and minimizing energy consumption.

Many algorithms have been developed to address the path planning problem, such as those based on A* [1-4] and Voronoi diagrams [5-8]. Path planning for AMRs faces many challenges in dynamic environments, because it is difficult to ensure the efficiency and safety under the obstacles and changing surroundings. Traditional path planning algorithms, such as A* and Dijkstra's algorithm, assumes the static environment typically. However, obstacles and goals may change at any moment in dynamic environments and the computed paths would be invalid. Many researchers have proposed path planning methods for dynamic environments; the dynamic A* algorithms can evaluate paths when obstacles change and utilizes sensors to acquire real-time environmental information, update obstacle positions, and adjust paths accordingly [9]. Liu et.al [10] proposed a dynamic fusion pathfinding algorithm to improve the calculation efficiency with the changes in the environment. An improved genetic algorithm-based dynamic path planning method enhances the efficiency and

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feasibility of path planning with incorporating visualization graphs, safety coefficients, floating-point encoding, hill-climbing methods, and particle swarm optimization rules [11].
Voronoi diagrams is popular methods to be used for path planning problems due to their ability to generate paths that maximize safety and clearance from obstacles. Despite the advantages of Voronoi diagrams, traditional Voronoi path planning methods have several challenges due to the real-time applicability in dynamic and constrained environments. The efficient skeleton extraction is a fundamental step in Voronoi-based navigation [5-8]. The concept was demonstrated the effectiveness in generating obstacle-avoiding paths by leveraging the inherent Efficient skeleton extraction in Voronoi-based navigation [8]. They proposed a connectivity-based and boundary-free skeleton extraction method to employ a distributed algorithm to connect identified skeleton nodes and generate a Voronoi diagram [8]. Due to the safety advantages using Voronoi diagram methods, the path planning approaches based on generalized Voronoi diagrams have many advancements. [12] proposed a path planning approach using generalized Voronoi diagrams to employ a parallel refinement algorithm to generate collision-free roadmaps and utilizes path planning to avoid congestion effectively.
Therefore, the development of the AMR system with high mobility and high navigation accuracy has become an important direction in smart factories. To make the AMR have good mobility and flexibility, the Mecanum Wheel with omnidirectional movement characteristics was used as the mobile platform. The Mecanum wheel is designed with a special wheel arrangement, which enables the robot to move in all directions without steering, including complex paths such as front and back, left and right and rotation. It can effectively cope with application scenarios such as limited space or dense obstacles in the factory [13]. After achieving omnidirectional mobility, how to effectively navigate and plan paths in complex environments has become one of the key technologies for successful AMR deployment.
In this study, the A* and Voronoi diagrams are used as the global path planning method, which can find the shortest path from the starting point to the end point in the known map. To use this path as the basis for AMR movement, it can effectively improve the navigation efficiency and accuracy. In practical applications, the AMRs are often affected by factors such as uneven ground and friction. They cause the problems such as positioning errors and the target offset. In order to solve this challenge, this study combines a depth camera and a visual odometer to fuse the information of the IMU (Inertial Measurement Unit) and the wheel speed odometer through the Kalman filter to improve the positioning accuracy and stability. and realize the position compensation and correction during slippage.

2. AMR design and control method

2.1. Model description of the AMR with the mecanum wheels

In this study, we used the AMR (AMR150m, produced by AUO) equipped with four Mecanum wheels, enabling omnidirectional movement, as illustrated in Figure 1. The design allows the AMR to manoeuvre in any direction without changing its orientation. The AMR is equipped with two LiDAR sensors (Orbbec MS500) to measure the environment's distance. The Orbbec MS500 uses innovative direct Time of Flight (dToF) ranging technology; it comprises a precise rotating-mirror optical scanning system and a high-frequency laser pulse transmitter. The LiDAR sensors are specifically designed for indoor mobile robotics and are integral to implement the Simultaneous Localization and Mapping (SLAM) algorithm. The LiDAR sensors allow the AMR to create detailed maps of its environment and navigate efficiently. For processing and control, the AMR uses a PC running Ubuntu 20.04 (Focal Fossa) and ROS Noetic.

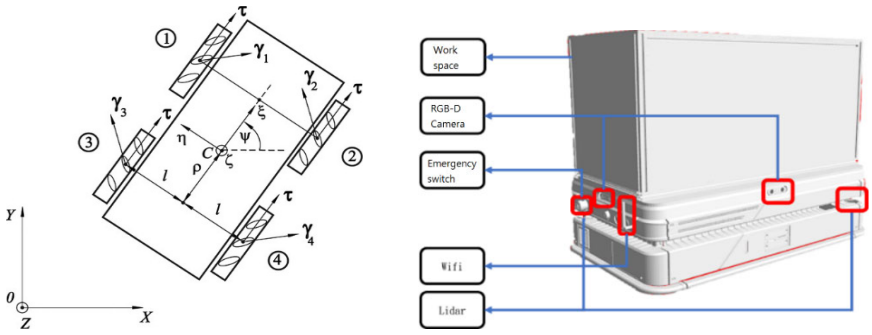


Figure 1. Four wheeled mobile robot with mecanum wheels 0.

2.2. Experimental system

The AMR system used in this paper is divided into five parts: the robot, mapping, positioning, navigation and data fusion; the AMR software architecture is shown in Figure 2. The Odometry, LiDAR, the depth camera and the inertial measurement unit (IMU) on the AMR provide the algorithm data and data fusion required for navigation. The localization maps are built by using GMapping, which uses the Odometry and LiDARs to build maps in indoor spaces for navigation and positioning modules. The positioning module uses the AMCL (adaptive Monte Carlo localization) algorithm to locate the AMR using the LiDAR and the map information. The autonomous navigation uses the extended Kalman filter (EKF) localization to fuse data. It corrects the problem of inaccurate movement caused by the slippage and add the fusion results to the Move-base module for navigation. The move-base module is shown in Figure 3 [14].

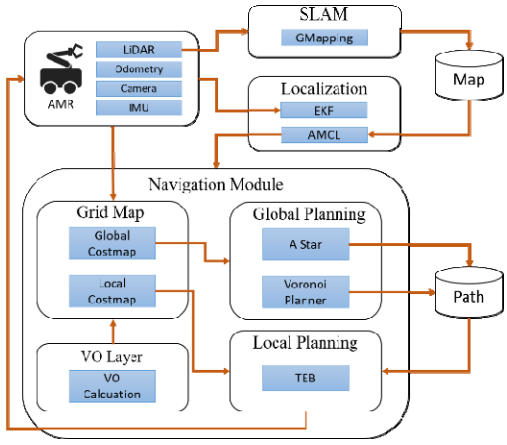


Figure 2. Software architecture of the AMR.

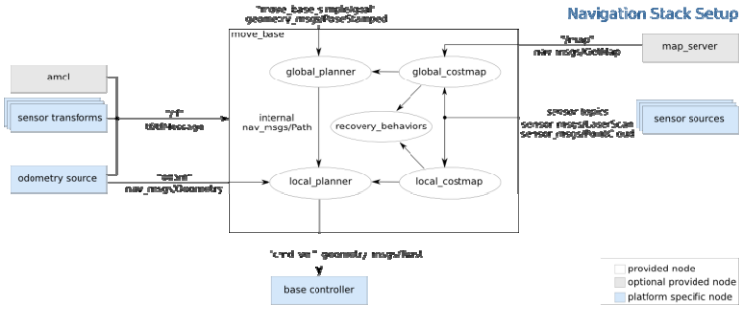


Figure 3. Move-base module [14].

2.3. Global path planning

In the robot navigation, the navigation requires the construction of an environment map through LiDAR and RGB-D sensors and SLAM (Simultaneous Localization and Mapping) technology. This map is usually stored as a Static Map or Global Map as the basis for subsequent route planning. The goal of Global Path Planning is to plan the best path from the starting point to the target point based on the environment map and ensure that the robot can avoid obstacles. There are many global planning algorithms, including A* algorithm [1-4], Dijkstra algorithm [3], Voronoi algorithm [5-8], Best-First Search (BFS) [15], and Rapidly-exploring Random Tree (RRT) [15]. In this study, we will use the A* algorithm and the Voronoi algorithm to study the differences.

(1) A* algorithm

A* algorithm is a heuristic path search algorithm, which widely used in the field of path planning. It starts from a specific starting node to find a path to the given goal node having the smallest cost. At each iteration of its main loop, A* needs to determine which of its paths to extend and an estimate of the cost required to extend the path all the way to the goal. A* selects the path that minimizes

$$f(n) = g(n) + h(n) \tag{1}$$

where n is the next node on the path, $g(n)$ is the cost of the path from the start node to n, and $h(n)$ is a heuristic function that estimates the cost of the cheapest path from $\boldsymbol{n}$ to the goal. The heuristic function is problem-specific and admissible. It never overestimates the actual cost to get to the goal. Then, A* is guaranteed to return a least-cost path from start to goal [16].

(2) Generalized voronoi diagram

Generalized Voronoi diagram (GVD) is a space segmentation algorithm used in path planning. [5] For over four decades, the Voronoi diagram can be traced back to the work of Rene Descartes in 1644 when he used a Voronoi-like diagram to cite the location of matter within the solar system. The researchers have developed many algorithms to compute the Voronoi diagram [5]. It presented work on a Voronoi diagram as an aid in robot route planning. To create the generalized path corridor based on the global path, the GVD is extracted from the grid map with the Voronoi-based parallel thinning (VPT) algorithm [17]. The GVD can be considered a collection of medial lines between the obstacles to provide a simple description of the environment, as shown in Figure 4.

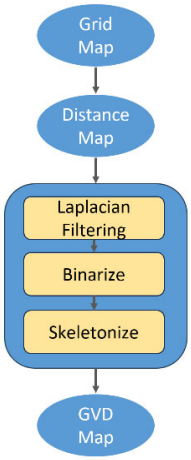


Figure 4. Generalized voronoi diagram.

As Figure 4 shows, the original map will be transferred by Laplacian filtering, binarize, and skeletonize process to transfer the skeletonized image. As shown in Figure 5, Figure 5(a) is the

original grid map representing the actual environment; Figure 5 (b) is called a distance map, where the grid map is transformed into a clearance map at first with the Euclidean distance transformation (EDT) algorithm [18]. Each cell in the clearance map comes with clearance value representing its distance to the nearest obstacle. Then the VPT is applied on the clearance map to extract the GVD of the grid map [shown as green line in Figure 5(c).

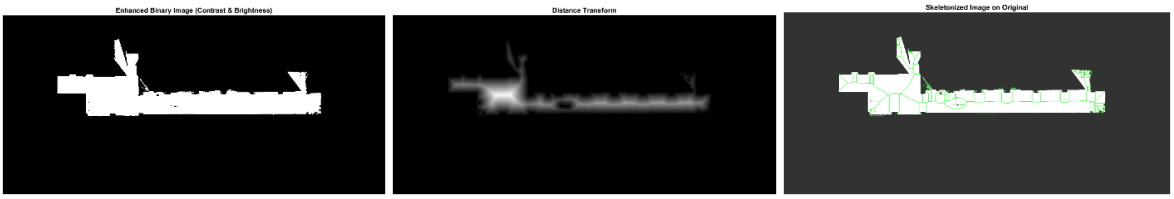


Figure 5. (a) Original grid map; (b) Distance transform; (c) Skeletonized image on original.

2.4. Local path planning

In a dynamic environment, the navigation based solely on global path planning cannot cope with sudden changes. Therefore, the robot needs local path planning to adjust the travel path in actual time. Local path planning requires robot to move in unknown environment or dynamic environment and it can be defined as real time obstacle avoidance by using sensory based information. The local path planning monitors the changes in the surrounding environment through LiDAR and RGB-D cameras and generates the short-term feasible paths to ensure that the robot can flexibly avoid obstacles and move forward smoothly. Local path planning is characterized by high flexibility and real-time performance. It can optimize paths based on real-time environmental information, allowing robots to navigate autonomously in complex scenarios.

In this study, the TEB (Time Elastic Band) is used as the local path algorithm for robot motion planning. The TEB is usually used in ROS for dynamic obstacle avoidance and path optimization of mobile robots. TEB treats path planning as an optimization problem through time parameterization, enabling the robot to flexibly adjust its path in a dynamic environment. [19] The core concept of TEB is the Elastic Band, which models the robot's initial path as a series of discrete poses and balances path length, smoothness, speed constraints, and distance to obstacles through mathematical optimization. Compared with the traditional DWA (Dynamic Window Approach), the TEB can generate smoother and more efficient paths through global optimization, which is particularly suitable for narrow spaces and dynamic environments. Since the TEB involves nonlinear optimization and has a large amount of computation, it may require additional parameter tuning to ensure performance in high-density obstacle environments.

3. Experimental results

The experimental design of this study is mainly aimed at comparing and analysing the performance of global path planning algorithms and positioning methods to verify the navigation accuracy and stability of the system in a real environment. The experiment is mainly divided into three parts: comparison of path planning methods, comparison of positioning methods, and measurement of visual positioning accuracy. A representative corridor-type field is designed in a simulated factory environment for testing. The appearance of the experimental field is shown in Figure 6.



Figure 6. Experimental field for the AMR.

The AMR navigation uses the motor encoders to record the distance and the angle of the robot, which can cause errors in the robot's mileage due to the external forces or the ground impacts. Therefore, some people use the vision to record the robot's mileage; they use camera images to estimate the device's movement trajectory. It extracts the image feature points to track their changes in continuous images; then, it calculates the camera's posture changes. However, this may cause problems with the odometer if the building features are too close. In order to correct the problem, a Kalman filter is used to predict and correct the coordinates.

This study used the robots_localization in ROS for the AMR's positioning via the fusion of multiple sensor data. It provides EKF to fuse data from IMU, Odometry, GPS, LIDAR, Visual odometry, etc. This study uses the odometry, the IMU and the visual odometry to perform data fusion. The experiment set fixed starting and target points in the same map environment to use A* and Voronoi algorithms for path planning respectively. The safety performance of the two methods is evaluated by comparing the navigation paths during actual navigation; then, we adjust the parameters of the expansion layer during navigation to observe the differences between the two methods. In the experiments, three positioning methods are used to verify the effectiveness of the positioning correction method proposed in this study.

Traditional wheel odometers (Odometry) rely solely on the robot's wheel speed to calculate its position, which is prone to accumulated errors due to tire slippage or uneven ground. Visual Odometry uses a depth camera to capture images and estimates the movement distance through changes in image feature points, thereby improving the ability to adapt to sliding or environmental changes. Sensor fusion positioning (Visual-Inertial Odometry with Kalman Filter) integrates visual odometer, IMU and traditional odometer information, and performs error correction and dynamic estimation through Kalman filter to improve overall positioning stability and accuracy. With Comparison of the experimental results, the actual navigation accuracy under different positioning methods is measured by comparing the fixed point markers with the true value coordinates. The program will subscribe to ROS Master through the Matlab GUI to receive map information. Then, it publishes the coordinates and directions of the target point to move_base for navigation and then subscribe to the coordinates of the base_link. The base_link coordinates are calculated by the relative relationship between the map and odometry. Then, the relative position of the robot in the map is sent to Matlab for receiving. The experimental data is transferred as shown in Figure 7 and the architecture of the AMR is shown in Figure 8.

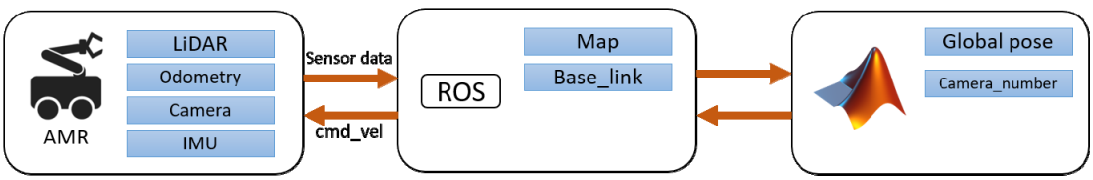


Figure 7. Sensor data for the AMR in Matlab GUI.

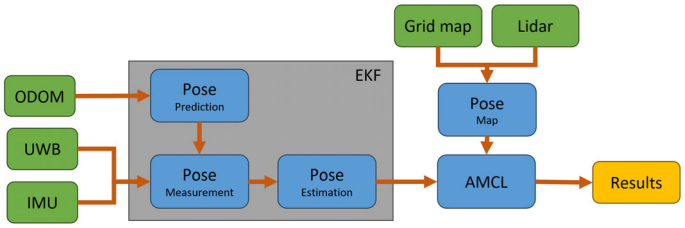


Figure 8. Architecture of the AMR using the AMCL with LiDAR.

3.1. Vision guidance

This experiment will test the front, left, and right depth cameras on the AMR in a spacious field and the field of view is shown in Figure 9. The distance between the robot and the tag is calculated by subscribing to the robot's odometer coordinates and the distance between the tag and the camera to calculate the relative address of the robot and the tag in the map. Assume that the robot coordinates are X0 and Y0, and the tag coordinates are X1 and Y1. Based on the coordinate diagram shown in Figure 10, the angle Yaw between the robot and the tag can be calculated using trigonometric functions, as shown in Eq. (2). After correcting the robot's posture so that it is parallel to the tag, the PID controller is corrected for left and right shifts based on the distance between the tag and the camera to complete the station entry guidance function.

$$Yaw = \tan^{-1}\left(\frac{Y_1 - Y_0}{X_1 - X_0}\right) \tag{2}$$

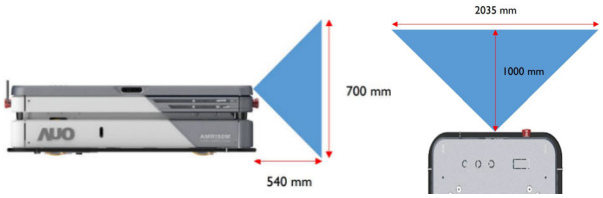


Figure 9. Field of view for the AMR.

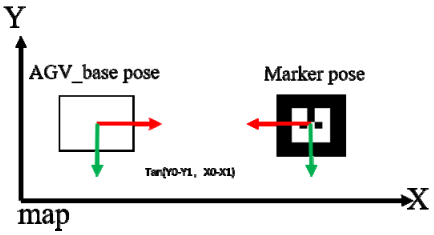
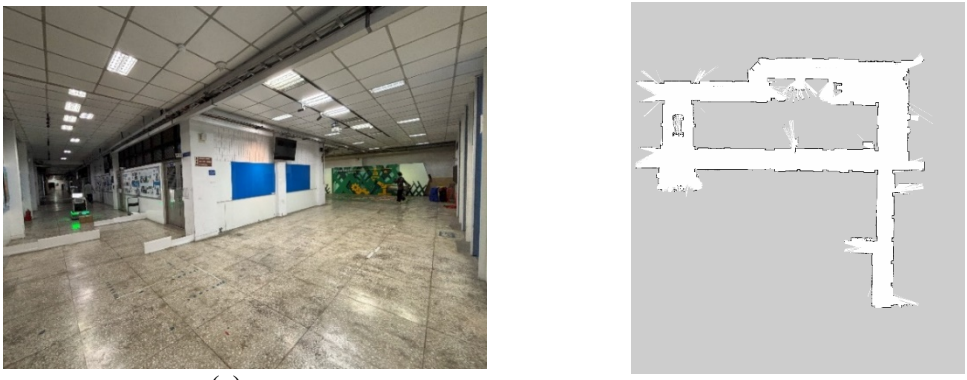


Figure 10. Coordinate diagram.

3.2. Results of global path planning

To evaluate the performance of the proposed method in the real-world environment, the environment was constructed as shown in Figure 11 (a) and the corresponding grid map was generated (as shown in Figure 11(b)). To validate the performance of the proposed method, the proposed algorithms are compared with A* algorithm and the Voronoi algorithm. First, the A* method is used for the

navigation with the inflation radius of 0.2, 0.5 and 0.8; the experiments of binary occupancy grids are shown in Figure 12. Observing the robot's navigation in Rviz with an inflation radius of 0.2m, it's clear that the route closely follows the obstacle. Due to the TEB (Time Elastic Band) characteristics of the regional routing algorithm, the robot avoids collisions with obstacles. With an inflation radius of 0.5m, when the robot stays slightly away to the obstacle than the case of the inflation radius of 0.2m. In order to check the A* method for the global path planning method, the navigation route of the inflation radius 0.5m is shown in Figure 13. The results can prove that the paths planned by A* algorithm can achieve the obstacle avoidance.



(a) (b)

Figure 11. The real environment to test the AMR with the A* and Voronoi algorithms.

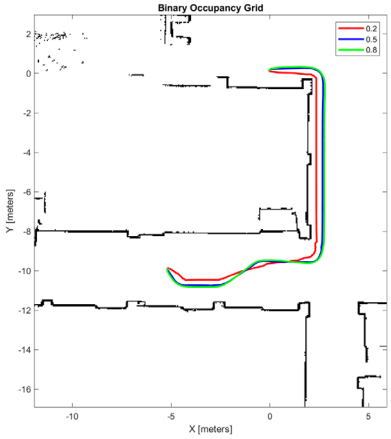


Figure 12. A* experimental results for inflation radius of 0.2, 0.5 and 0.8.

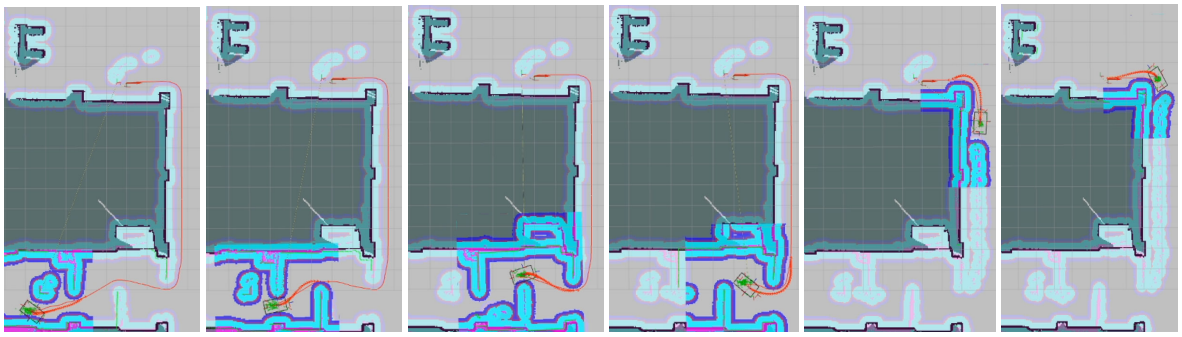


Figure 13. Comparison of the A* algorithm (the inflation radius is 0.5).

To compare the trajectories produced by A* algorithm and Voronoi algorithm, Figure 14 shows the results of the global path planning for Voronoi algorithm with the inflation radius to 0.2m, 0.5m, and 0.8m. Figure 14 shows that the global path planning using the Voronoi algorithm is not affected by the inflation radius. Figure 15 shows the path obtained by Voronoi planner; the guidance result is distant from obstacles. The Voronoi algorithm has an advantage over A* in terms of path smoothness, but the moving time is longer than the A* algorithm, as shown in Table 1.

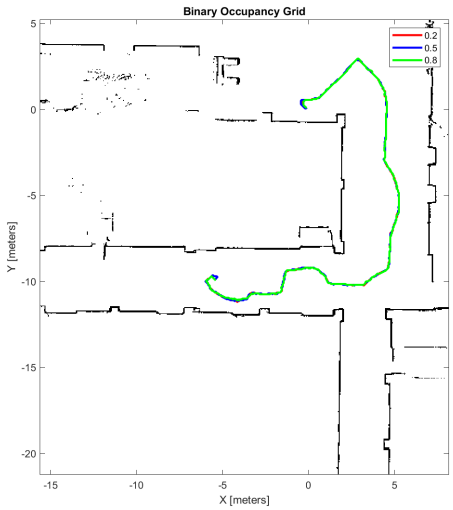


Figure 14. Voronoi experimental results.

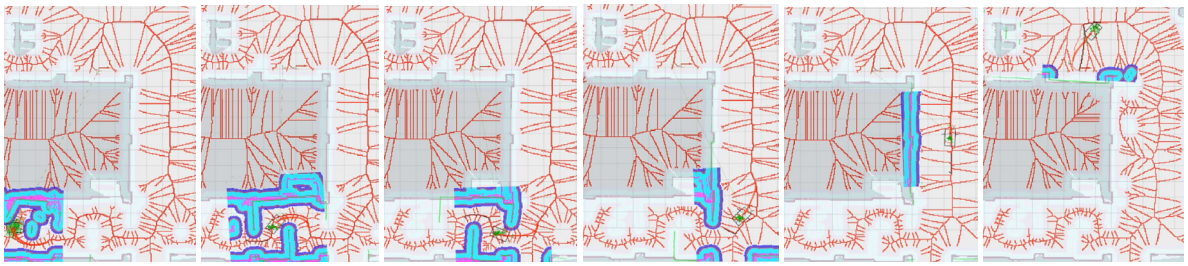


Figure 15. Comparison of the Voronoi algorithm (the expansion layer is 0.5).

Table 1. The moving time versus A* algorithm

Inflation radius	0.2m	0.5m	0.8m	Average time(sec)
A* algorithm	89	87	92	89.33
Voronoi algorithm	112	109	114	111.67

3.3. Data fusion navigation accuracy measurement

This section presents a comparative experiment on global path planning with using the Voronoi diagram. The data fusion parameters are mainly based on the visual odometry. Traditional odometry mainly considers the direction and rotation angle of the selected corner, while IMU mainly considers the linear velocity in the X direction and the angular velocity of the selected corner; the experimental results are shown in Tables 2-4. After the data fusion processing, the navigation accuracy has been improved. The error in the X-direction distance has been reduced by about 6 centimeters, and the error in the Y-direction has been reduced to less than 1 centimeter (0.76 cm). The center offset is also a relatively good result after data fusion. In Table 3, the error by the visual odometry is as follows,

Error_x=9.7 (cm), Error_y=3 (cm), and Error_yaw=0.2˚. In Table 4, the error by the traditional odometry is as follows, Error_x=12.96 (cm), Error_y=3.96 (cm), and Error_yaw=0.37˚.

Table 2. Error by the data fusion.

	X(m)	Y(m)	Yaw (˚)	Error
Target	6.269	0.614	0.14	
1	6.263	0.612	0.34	0.006325
2	6.331	0.631	0.29	0.064288
3	6.318	0.609	0.17	0.049254
4	6.39	0.613	0.43	0.121004
5	6.346	0.627	0.34	0.07809
RMSE	0.063	0.0076	0.174	0.0638

Table 3. Error of the visual odometry.

	X(m)	Y(m)	Yaw (˚)	Error
Target	6.269	0.614	0.14	
1	6.249	0.576	0.22	0.042942
2	6.317	0.584	0.3	0.056604
3	6.317	0.642	0.37	0.05557
4	5.991	0.578	0.41	0.280321
5	6.362	0.633	0.44	0.094921
RMSE	0.0974	0.0302	0.208	0.106

Table 4. Error of the traditional odometry.

	X(m)	Y(m)	Yaw (˚)	Error
Target	6.269	0.614	0.14	
1	6.477	0.667	0.2	0.214646
2	6.412	0.658	0.37	0.149616
3	6.379	0.656	0.28	0.117745
4	6.265	0.642	0.75	0.028284
5	6.452	0.645	0.95	0.185607
RMSE	0.1296	0.0396	0.37	0.1392

4. Conclusions

This study successfully designed and implemented an autonomous mobile robot (AMR) system with omnidirectional mobility, high navigation accuracy, and stable docking capabilities. It integrated the AMR with the Mecanum wheels for the A* and Voronoi global path planning algorithms. The TEB regional path planning, vision and inertial sensor fusion positioning technology, and AR marker-based vision guidance are used for the AMR to improve the autonomous navigation and handling efficiency in smart factories. Experimental results show that the Voronoi algorithm performs better than A* in

obstacle avoidance and safety. Voronoi algorithm is more suitable for use in factory environments with obstacles. Data fusion technology significantly improves positioning accuracy through the integration of visual odometers, IMUs, and traditional odometers. The offset error is reduced to less than 6 centimeters. In terms of visual guidance, the average positioning error of the AMR in three directions is under 2 centimeters through PID control combined with AR marker recognition. They indicate that the AMR system has good alignment capabilities and the proposed method has high mobility, high accuracy and high stability. It can effectively cope with the complex and dynamic environmental requirements in the factory and It can be further applied to various automation fields such as smart manufacturing, warehousing and logistics in the future.

This study also found several potential optimization areas: in terms of visual guidance, deep learning image recognition models can be further introduced to improve recognition accuracy under different lighting or complex backgrounds. The existing system mainly uses static maps for navigation. In the future, it can be considered to integrate dynamic map construction and real-time obstacle recognition technology to improve responsiveness in dynamic environments. Furthermore, the positioning module can add high-precision positioning technologies such as UWB or RTK GPS to compensate for positioning problems which the visual features cannot be effectively obtained. Finally, the system can also consider adding a multi-machine collaborative architecture to further develop group AMR collaborative handling technology to cope with more diverse and large-scale logistics needs.

Acknowledgement

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